MIL-E-1/1536A
4 January 1971
SUPERSEDING
MIL-E-1/1536
1 July 1964

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NUMERICAL INDICATOR

TYPE 8502

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Gas filled, glow discharge (neon), cold cathode

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ebb	Ik(individual)	Rp	TE	Alt
Unit:	Vdc	mA	Ohms	°C	ft
Maximum:	---	2.0	---	+85	70,000
Minimum:	120	---	---	-65	---
TEST CONDITIONS:	120	---	20K ±1%	---	---

GENERAL:

Qualification - Required

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:55Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Qualification inspection							
1006	Salt spray (corrosion)		---	---	---	---	---	---
	Quality conformance inspection, part 1							
3347	Ionization voltage (1) (with illumination)	Ambient illumination = 5 to 50 foot candles	---	---	Ebb	---	120	Vdc
3333	Uniformity of cathode glow	Ambient illumination = 5 to 50 foot candles	---	---	---	---	---	---
1256	Electrode current (cathode)	Individual cathodes	---	---	Ik	0.7	1.4	mA
	Quality conformance inspection, part 2							
1002	Barometric pressure, reduced	eb = 120; Rp = 20 K; pressure = 35 mmHg (see note 1)	---	---	---	---	---	---
1031	Variable-frequency vibration	See notes 2 and 3	---	---	---	---	---	---
---	Light output	Light individual cathode (see notes 4 and 10)	---	---	---	0.10	---	ftC
---	Alinement digits	See note 5	---	---	---	---	±5	Degrees
---	Shock	250 G; duration = 1.0 ±0.5 ms (see notes 2 and 6)	---	---	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions (1)	Group C; sequential counting; 1 Hz; t = 1,000 hours (see note 7)	---	---	---	---	---	---
---	Life-test end points (1)	Uniformity of cathode glow	---	---	---	---	---	---
		Cathode current	---	---	Ik	0.5	1.9	mA
		Light output	---	---	---	0.1	---	ftC
---	Life-test provisions (2)	Group C; one numeral cathode glowing steadily; t = 500 hours (min) (see notes 8 and 9)	---	---	---	---	---	---
---	Life-test end points (2)	Uniformity of cathode glow, numeral under test	---	---	---	---	---	---
		Cathode current, numeral under test	---	---	Ik	0.5	1.9	mA
		Light output, numeral under test	---	---	---	0.1	---	ftC

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:55Z
Check the source to verify that this is the current version before use.

NOTES:

1. The tube shall be tested by using a 60-Hz sine-wave full-wave rectified and unfiltered, having the indicated peak value.

2. No permanent shorts or opens shall be permitted.

3. On each plane the 10 cathodes shall be sequentially energized at the test conditions.

4. Light output of each numeral is measured with a foot-candle (ftC) meter and an eye-corrected photronic cell. The entire cell area is illuminated and is coupled to the tube by a light-tight cylinder so that the face of the cell is 1-1/2 inches (38.10 mm) from the tube base.

5. Centerline of numerals, drawn through the top of the numeral support posts, shall be within the tolerance specified in relation to line drawn on the base of the tube through pins 1 and 7.

6. Shock test shall be made on any shock machine capable of producing a half-sine wave-shock waveform of the specified duration and amplitude. No voltages shall be applied during this test. Each tube shall be subjected to a total of 20 shocks, i.e., 5 shocks in each of positions (X, Y, Z plus, and Z minus) in any sequence. This is a destructive test. Tubes subjected to destructive tests shall not be delivered on the contract or order.

7. Tubes shall be energized with a duty cycle greater than 90 percent.

8. Group C, life-test sampling plan, and Group C, establishment of prerelease eligibility (MIL-E-1 paragraphs) should be used with the following exceptions:
   (a) Group C, life-test sampling plan:
      (1) The first sample size = 3.
      (2) The first cumulative sample size = 3.
      (3) The second sample size = 3.
      (4) The second cumulative sample size = 6.
   (b) Group C, establishment of prerelease of eligibility:
      (1) A minimum of 15 tubes shall - - - - - - - - etc.
      (2) Replace all reference to the current 4-tube sample with current 3-tube sample. Replace all reference to the last 16 with the last 15.

9. One tube shall have the top numeral cathode glowing steadily. One tube shall have the bottom numeral cathode glowing steadily. One tube shall have the "0" numeral cathode glowing steadily.

(A) 10. After a quiescent state of two or more weeks, the numerals, if required, shall be given a 90-second warmup prior to checking for full glow.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy -
Air Force - 11, 17, 80
DSA - ES

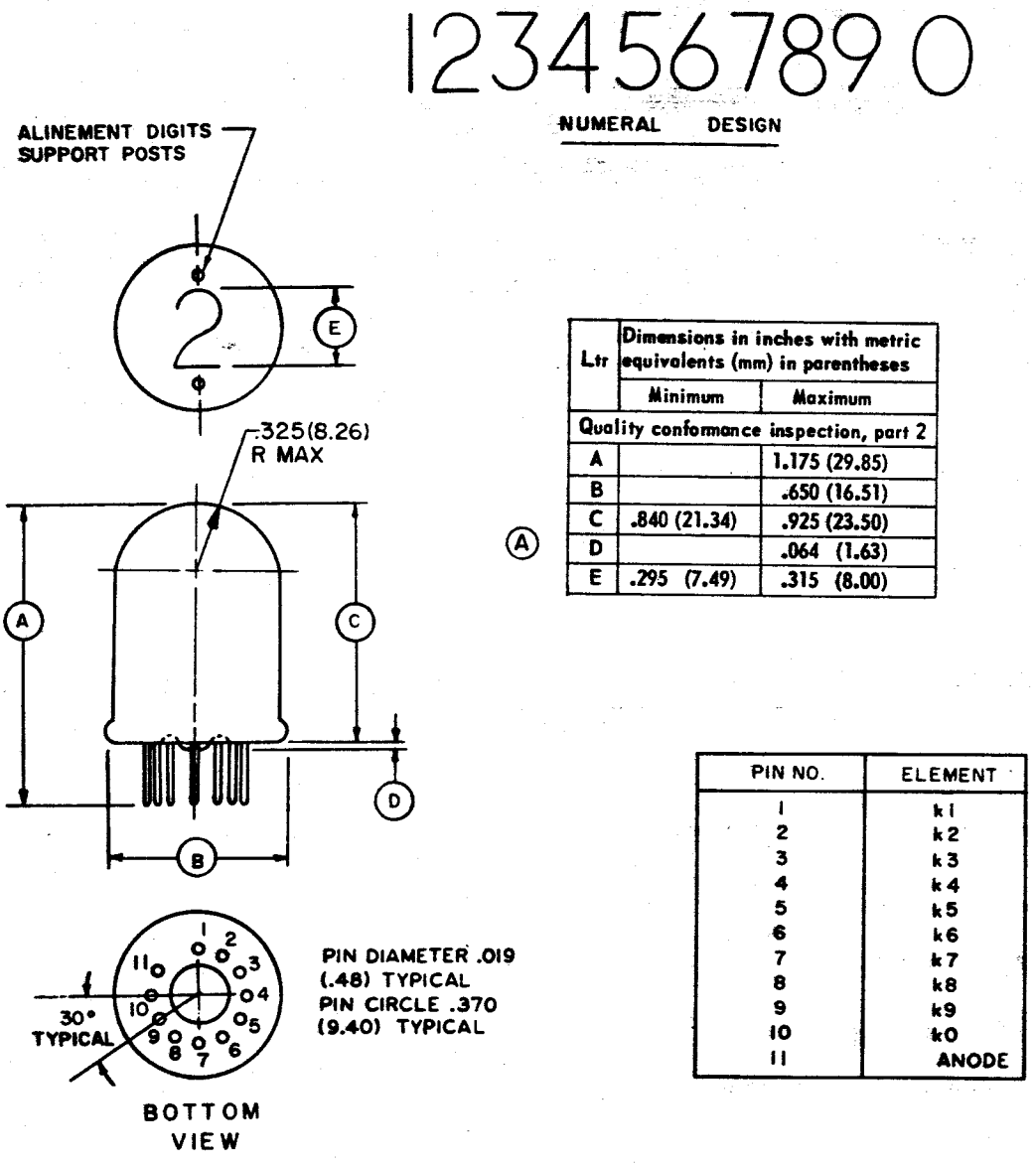
User activities:
Army - WC
Navy - AS, OS, CG, SH
Air Force - 19

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2425-17)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:55Z
Check the source to verify that this is the current version before use.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		1.175 (29.85)
B		.650 (16.51)
C	.840 (21.34)	.925 (23.50)
D		.064 (1.63)
E	.295 (7.49)	.315 (8.00)

PIN NO.	ELEMENT
1	k1
2	k2
3	k3
4	k4
5	k5
6	k6
7	k7
8	k8
9	k9
10	k0
11	ANODE

FIGURE 1. Outline drawing of electron tube type 8502.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:55Z
Check the source to verify that this is the current version before use.